

Physics-Informed Neural Networks on Effect of Temperature Induced Excess Pore-water Pressures on the Shaft Bearing Capacity of Geothermal Piles

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1 Introduction

Geothermal piles have a double function. On the one hand, they operate as traditional foundation elements. On the other, they are used to exchange heat from the ground for heating and cooling of superstructures. In their cooling mode, the temperature of the circulated fluid is higher than the soil's temperature; hence, increasing the temperature of the latter. Under normal operating conditions, the fluid can be up to 30°C, although greater temperatures have been tested. In low permeability soils, these temperature increases have the potential to increase the pore-water pressures and reduce the available effective stress. If this reduction is in the same order as the mobilised shaft friction between the pile and the soil, their effect on the shaft resistance can be significant and reduce the ability of the pile to carry the intended loads (Fuentes et al. [2016]).

In the presence of lower permeability soils, these excess pore-water pressures can reach values in the order of 1 MPa for temperature increments of 30°C, which in most practical cases of bearing piles would exceed the effective stress at the interface.

2 Objective

In this study, we want to use Physics Informed Neural Networks (PINNs) to solve the fully coupled formulation governing the development of excess pore-water pressures in geothermal piles and its impact on the shaft friction at the pile–soil interface. The emphasis should be on presenting comparisons in terms of orders of magnitude of excess pore-water pressures and not attempting to specify accurately all properties, as this will change from case to case.

Fig. 1 shows the problem's geometry. A single pile diameter equal to 1 m and pile length of 25 m were used. This length is enough to guarantee that seasonal effects are less important at mid-depth of the pile where the comparison between methods is carried out.

The problem presents geometrical axisymmetry about the pile's axis, so a cylindrical coordinate system (r, θ, z) was chosen as shown in Fig. 1. Additionally, the temperature difference at the pile surface for different positions within a pile diameter is lower than 2°C; therefore, the azimuthal coordinate θ can be eliminated. Likewise, it is assumed that the temperature of the pile along its length is constant. This, combined with an assumption of fully hydrostatic initial pore-water profile, allows eliminating the z coordinate as well. The problem then becomes one-dimensional, defined in the radial direction, r .

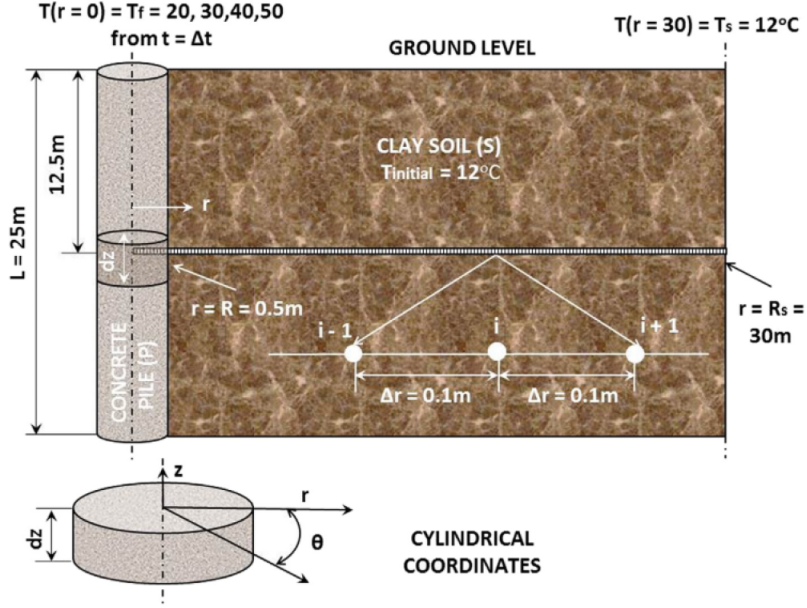


Figure 1: Problem definition.

3 Governing Equations

- The heat equation for a constant thermal conductivity applies to both the pile and soil, and is:

$$\frac{\partial T}{\partial t} = \alpha \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) = \alpha \left(\frac{1}{r} \frac{\partial T}{\partial r} + \frac{\partial^2 T}{\partial r^2} \right) \quad (1)$$

- The combination of soil and water mass balance formulations yields the final governing second-order parabolic differential equation that applies only to the soil mass:

$$-[(1-n)\beta_s + \beta_w n] \frac{\partial T_s}{\partial t} + \left(n\alpha_w + \frac{1}{K_s} \right) \frac{\partial u}{\partial t} - \frac{k}{\lambda_w} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) = 0 \quad (2)$$

where the unknowns are the soil temperature T_s and the excess pore-water pressure u .

- The following boundary conditions are applied:

$$\left(\frac{\partial u}{\partial r} \right)_{r=R} = 0 \quad ; \quad u(r = R_s) = 0 \quad \text{at any time} \quad (3)$$

$$u(t = 0) = 0 \quad \text{throughout} \quad (4)$$

$$T(r = 0.5) = T_f \quad ; \quad T(r = R_s) = T_s \quad (5)$$

$$T(t = 0) = T_s \quad \text{throughout} \quad (6)$$

Table 1: Parameter values for the parametric study.

Variable	Soil particles (P)	Water (w)	Soil medium (S)
Porosity, n	—	—	0.25
Thermal expansion coefficient, β (1/°C)	3.00 E−05	3.42 E−04	1.10 E−04 ^a
Compressibility constant, water, α_w (1/Pa)	—	5.00 E−10	—
Density, ρ (kg/m ³)	2700	1000	2275 ^a
Thermal conductivity, Γ (W/m °C)	—	—	2.0
Specific heat, C (J/kg °C)	837.2	4186	1674.4 ^a
Permeability of Soil, k (m/s)			
1.00 E−8 1.00 E−9 1.00 E−10 1.00 E−11 1.00 E−12			
Soil compressibility, K_s (Pa)			
2.00 E06 2.00 E07 2.00 E08 2.00 E09 2.00 E10			
Temperature of the fluid, T_f (°C)			
20 30 40 50			

^a These variables were calculated using the rule of mixes—e.g., for density, $\rho_s = \rho_P(1 - n) + \rho_w n$.

4 Tasks

In this project, Physics-Informed Neural Networks (PINNs) are used to model the coupled thermo-hydro behavior of geothermal piles. The key tasks are outlined as follows:

1. Nondimensionalization of Governing Equations

Reformulate the coupled heat conduction and pore-pressure equations into nondimensional form to improve numerical stability and generality of the PINN models.

2. Forward PINN Model

Develop a Physics-Informed Neural Network to solve for the temperature and excess pore-water pressure fields for the specific case:

$$T_f = 50^\circ\text{C}, \quad T_s = 12^\circ\text{C}$$

Only the initial and boundary conditions should be used as supervised loss terms—no labeled data from the interior of the domain.

3. Inverse PINN Model

Implement an inverse PINN to estimate the permeability k , using data from a different case than in the forward model. The goal is to learn the material parameter based on observed temperature and pressure fields.

4. Bonus: Parametric PINN Model

Extend the PINN architecture to include k as an input variable. Use a fixed soil compressibility:

$$K_s = 2 \times 10^6 \text{ Pa}$$

Evaluate the model for the forward case $T_f = 50^\circ\text{C}$, $T_s = 12^\circ\text{C}$, with a range of permeability values:

$$k \in \{10^{-8}, 10^{-9}, 10^{-10}, 10^{-11}, 10^{-12}\}$$

For each permeability configuration, reference solutions are provided as a dataset, pre-computed using a Finite Difference Method (FDM) solver. These datasets are used to benchmark the predictions from the PINN.

As before, the temperature field inside the pile is not modeled. A constant temperature is applied at the soil–pile interface, simplifying the thermal boundary condition.

5. Presentation

Prepare a concise presentation including:

- Nondimensionalization process and governing equations
- Forward and inverse PINN architecture and training
- Parametric study setup and evaluation
- Comparison with FDM results and error analysis

5 Reference

Raul Fuentes, Nuria Pinyol, and Eduardo Alonso. "Effect of temperature induced excess porewater pressures on the shaft bearing capacity of geothermal piles". *Geomechanics for Energy and the Environment*, 8:30–37, 2016.